



124105093

DEPARTMENT OF MATHEMATICS NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA

B.Tech. Second Semester (EE/CE/ECE/ME/RA/SET)

Mid Term – I Examination

Date: February 20, 2025

Max. Marks: 15

Time: 9:00 am – 9:50 am

Paper: Integral Calculus and Difference Equations (MAIC 102)

Instructions:

- Answer all the questions.
- The calculator is not allowed into the examination hall.
- Start writing your answers from the first page. Do not leave it blank. No extra answer sheets will be provided.

Q1. Find a power series solution of $x(1+x)\frac{d^2y}{dx^2} + (x+5)\frac{dy}{dx} - 4y = 0$ about $x = 0$. (5M)

Q2. Find all the eigenvalues and eigenfunctions of the Sturm-Liouville boundary value problem $y'' + \mu y = 0$, $y(0) = 0$, $y\left(\frac{\pi}{2}\right) = 0$. (5M)

Q3. Evaluate $\int_0^1 \int_{x^2}^{2-x} xy \, dy \, dx$ by changing the order of integration. (5M)